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Chapter: Integrated Pest Management on Mustard Crop

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Indian vegetable oil economy is the world's fourth largest after United States, China and Brazil. India is third largest rapeseed-mustard producer in the world after China and Canada with 12 per cent of world total production. Rapeseed-mustard ranks second after soybean among edible oilseeds in production and contributes to more than 30% of edible oil production in India. Rapeseed-mustard group of crops comprise mustard/raja, toria, brown Sarson, Yellow Sarson, Gobhi Sarson, black mustard and taramira. The crops occupy an area of 6.01 million hectare, yield 8.04 million tons with average productivity of 1339kg/ha (AICRPRM, 2018). Mustard is mainly cultivated in India, which contributes about 90 per cent of the total rapeseed-mustard production. It is predominately grown in Rabi season in wheat belt of northern (Haryana, Punjab and Jammu and Kashmir) India.

The important factors causing low and fluctuating production of mustard in India are low or non-adoption of package of improved production technology, susceptibility of mustard varieties to pest and diseases and non-adoption of pest and diseases management practices. The major insect-pests of mustard include Aphid (*Lipaphis erysimi*) and Painted bug (*Bagrada hilaris*), threatening the crop right from sowing until end of crop season. Among diseases white rust (*Albugo candida*), Sclerotinia rot (*Sclerotinia sclerotiorum*), Alternaria blight (*Alternaria brassicae*) and powdery mildew (*Erysiphe cruciferum*) are the major diseases, which reduced the yield potential of rapeseed mustard substantially.

#### KEY PESTS OF MUSTARD

**Mustard Aphid (*Lipaphis erysimi*):** This is a major pest of mustard. It causes loss to the crop from December to March. This insect can cause loss from 25 to 40 per cent. Both nymphs and adults suck cell sap from leaves, stems, inflorescence and developing pods, as a result plants remain stunted, reduced pod and grain number, pod shrivel up and seed do not develop. Economic threshold level of this pest is when pest population reaches 20-25 aphids/plant and when 30 per cent plants are infested.

**Painted Bug (*Bagrada hilaris/cruciferarum*):** This pest attacks the crop at two stages in the season i.e. at initial stage in October-November and crop maturity stage in March-April. The nymphs and adults suck cell sap from leaves and pods, which eventually wilt and dry up. This insect also reduced the oil quantity by sucking pods, pods shrivel up and seed do not develop.

## MAJOR DISEASES OF MUSTARD

**Alternaria blight:** In mustard crops, this disease is mainly caused by *Alternaria brassicae*. However, other species of *Alternaria* that is, *A. brassicicola*, *A. raphani* and *A. alternata* have also been reported parasitizing these crops in India. This is widely distributed, more destructive and most damaging disease under epiphytotic conditions, causing yield loss up to 70 per cent. Heavy losses in yield occur if favourable conditions like an average temperature of 18°C with relative humidity 80 percent or above and stormy weather prevail especially during flowering and pod development stages. The shrivelled and discoloured seeds fetch lower market price. The disease is characterized by formation of brown colour concentric rings on leaves, stems, siliquae and pods. *Alternaria brassicae* survives on diseased plant debris in the soil and many alternate cruciferous hosts like cauliflower, cabbage, turnip, radish etc.

**White rust:** This is a common disease of mustard caused by fungus *Albugo candida* and attacks all the plant parts except roots. The disease appears as prominent white creamy scattered raised and roundish pustules on the under surface of lower leaves. Many pustules coalesce and form large patches which cover entire lower surface of the leaf. This disease when appears on stag head phase distorts inflorescence, where it causes hypertrophy and hyperplasia causing 17-34 per cent loss in yield. White and creamy pustules also appear on the hypertrophied parts.

**Sclerotinia rot:** Sclerotinia rot caused by fungus *Sclerotinia sclerotiorum* is a major disease in mustard. Its infection at early stages of plant growth results in complete failure of the crop whereas late infection lowers the yield quantity and quality as well. Disease increases with mono-cropping of mustard. In individual affected plants sometimes no grain is formed. In recent years, Sclerotinia rot has emerged as a major problem in mustard in North India. The extent of damage is 40-80 per cent in Haryana and Punjab. The disease appears as elongated, buff to light brown water soaked lesions, which later rot and are covered with white, cottony mycelial growth of the fungus. All the affected parts of the plants rot in cool and wet weather. The affected plants show stunting and premature ripening, shredding of stem, wilting and drying. A large number of black sclerotia appear in fungal growth around the rotted stem. The sclerotia survive in the soil for longer period.

**Powdery mildew:** Caused by *Erysiphe cruciferarum*, this is a disease of warmer and drier tracts, where mustard is grown. The disease is gradually becoming common with shortening of winter and climate change. The disease usually arrives in later part of crop. However, it is also observed during vegetative stage, whereby the pathogen could cause significant loss to the crop.

**Broomrape (*Orobanchae aegyptica*):** Mustard crop in Haryana over large areas has been severely infested with holoparasitic weed broomrape, which has threatened cultivation in these areas. This parasitic weed grows on the roots of mustard plants in response to germination stimulants secreted by its roots. As infestation of this weed starts after 7-10 days of sowing of oilseed Brassica, control measures in early stages of crop growth should be applied.

## IPM INTERVENTIONS AT DIFFERENT GROWTH STAGES

### Pre-sowing stage:

- Deep summer ploughing the soil to expose soil-borne pathogens and destroy eggs of painted bug in order to reduce primary source of inoculum
- Preparation of level and well drained field to ensure proper drainage of water
- Follow appropriate crop rotation and balanced dose of fertilizers as per location-specific recommendation
- Sesbania green manuring along with soil incorporation of mustard waste @ 2.5 ton/ha in Kharif season
- Removal of pest debris and residue of previous crop to reduce soil-borne inoculum of diseases (painted bug also thrives on crop residues and weeds)

### Sowing stage:

- Soil incorporation of Trichoderma-based product @ 2.5 kg/ha pre-incubated in 50 kg of well-rotten farm yard manure to reduce soil-borne inoculum of diseases
- Sowing at proper time (01-31 October) escapes the attack of aphid, painted bug and white rust
- Use of disease-resistant hybrids and varieties recommended for the region
- Seed treatment with freshly prepared aqueous garlic bulb extract (1% w/v) or Trichoderma-based product @ 10 g/kg
- In case of downy mildew infection, the disease is brought under control by treating the seeds with metalaxyl-M 31.8% ES @ 6 ml/kg seed
- Avoid narrow spacing/heavy seed rate

### Seedling and vegetative stage:

- Maintain recommended spacing of plants by thinning of crop
- Irrigation of crop at seedling stage to protect against painted bug
- Maintenance of weed-free crop by clean cultivation
- Regular monitoring of crop and destroying of pest infested/infected plants
- Spray application of micronutrients like boron and zinc are also very useful in pest management
- Judicious use of irrigation depending on soil type and rainfall; irrigation after vegetative stage should preferably be avoided
- Hand picking of aphid-infested twigs in the initial attack
- Conservation of natural enemies of aphids namely *Coccinella septempunctata* (ladybird beetle), *Chrysoperla carnea* (green lacewing), *Syrphid* fly, etc.
- Application of two drops of soybean oil per young shoot of *Orobanche* reduces the infestation
- Removal of heavily diseased plants from the field and apply need-based spray of freshly prepared aqueous garlic bulb extract (2% w/v)

Flowering and pod formation stage:

- Regular monitoring of crop field
- Foliar spray of freshly prepared aqueous garlic bulb extract @ 1% (w/v) at early bloom
- If mustard crop is sown late and fertilized excessively with nitrogen, the crop tends to get affected more severely by diseases but can be protected from major diseases by spraying the crop at flowering-to-early pod formation stage with a mixture of metalaxyl 4% and mancozeb 68%

#### RAJASTHAN FIELD VALIDATION (SUCCESS STORY)

ICAR-NCIPM has worked for more than a decade synthesizing and validating IPM in mustard, demonstrated at village level in Haryana and Rajasthan in farmers' participatory mode. In 2010, after introduction of high-yielding hybrids by ICAR-AICRP on rapeseed-mustard, Sclerotinia rot disease emerged as a serious problem in mustard because some hybrids are highly susceptible to this disease.

A farmer of Siyali Khurd village (District Alwar, Rajasthan; latitude N27°54'23.2", longitude E76°36'27.7"), about 150 km from IARI Campus, Pusa, New Delhi, approached NCIPM because of severe incidence of Sclerotinia rot. Prior to IPM implementation, farmers of the area were taking monoculture of mustard without crop rotation and deep summer ploughing, and no seed treatment with Trichoderma was undertaken.

IPM practices implemented: recommended dose of fertilizer along with gypsum @ 250 kg/ha and Potash @ 40 kg/ha, soil incorporation of Trichoderma @ 2.5 kg/ha pre-incubated in 50 kg well-rotten FYM, and seed treatment with Trichoderma @ 10 g/kg seed. By destroying previous crop residue, the crop was protected from painted bug infestation and soil-borne pathogen-caused disease loss. Appropriate seed rate of 4 kg/ha was followed along with region-specific optimum sowing time, which helped escape incidence of aphids, Alternaria blight, white rust and Sclerotinia rot.

Based on this baseline (2011), the IPM module was fine-tuned and implemented across 40 ha in 2011-12 in Haryana and Rajasthan, expanding through 2012-14 as more farmers joined. Implementation resulted in significant reduction in severity of Sclerotinia rot with higher yield and benefit-cost ratio compared to farmers' conventional practices. After 2014, crop-stage-based IPM interventions were validated across 20 ha in farmers' participatory mode in districts of Alwar (Rajasthan), Mahindergarh-Jhajjar (Haryana), and Morena-Bhind (Madhya Pradesh), in collaboration with Krishi Vigyan Kendra Navgaon (Alwar) of Sri Karan Narendra Agriculture University, Jobner, Rajasthan.

**ECONOMIC IMPACT:** Crop stage-based IPM mustard module implementation across 100 ha in farmers' participatory mode (KVK Mahindergarh, KVK Jhajjar, KVK Navgaon-Alwar, 2014-2019) provided a 10% advantage over farmers' conventional practices. Each additional rupee invested in IPM technology adoption gave Rs. 5.1 in return.

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